# Practical Incentive-Compatible Drafts: Carry-Over Lottery Allocation

Paper Track: Basketball

Paper ID: 100

Open-source repository: https://github.com/kvr06-ai/cola-sloan-ssac27

## 1. Introduction

On February 7, 2026, in a National Basketball Association (NBA) game, the Utah Jazz led the Orlando Magic 94-87 going into the fourth and final quarter. The Jazz then kept two of their best players, Lauri Markkanen and Jaren Jackson Jr., out of the game for the whole quarter. Orlando outscored them 33-23 and won 120-117 [1]. Five days later, the NBA fined the Jazz $500,000 over that game and one other [2]. The Jazz had company. The Ringer, a sports website, counted eight of the league's thirty teams trying to finish with the worst records that month [3]. Losing on purpose in this way is called tanking.

Those teams wanted a better place in the draft, the yearly event in which NBA teams take turns choosing new players. Each turn is called a pick, and earlier picks give teams a better choice. In 2026, the available players were led by AJ Dybantsa and Darryn Peterson [3]. Sixteen of the thirty teams reach the playoffs, the tournament that follows the regular season and decides the champion. The fourteen teams that miss the playoffs pick first in the draft, and a lottery, which is a random drawing, decides which of them get the earliest picks. Under the lottery the NBA used from 2019 to 2026, the three teams with the worst win-loss records each had a 14% chance at the first pick. The team with the fourteenth-worst record had a 0.5% chance. The drawing decided only the first four picks. The rest went in reverse order of record, which means the worst remaining team picked next [4]. A worse record therefore meant better odds in the drawing and an earlier pick. Taylor and Trogdon [5] compared three NBA seasons played under different draft rules and found that teams lost more often when the rules rewarded losing.

The draft exists to help weak teams, and that goal creates the problem. At this conference in 2020, Banchio and Munro [6] studied lotteries that set their odds by the end-of-season standings, the final ranking of teams by record. They proved that every such lottery gives some team a reason to lose on purpose, unless it gives all the teams that miss the playoffs the same odds. A lottery of this kind therefore cannot both favor the worst teams and remove the reason to lose. We call this the impossibility result. Every change the NBA has made to its lottery has traded one of these two goals against the other. The latest, called 3-2-1, was adopted in May 2026. It gives the three teams with the worst records fewer chances in the drawing than the seven teams just above them. The teams at the bottom therefore no longer gain by losing [7]. A team in the middle of the standings still can, because of the play-in tournament. The NBA is split into two conferences, East and West. In each one, teams that finish seventh to tenth enter the play-in, a short qualifying round for the last playoff spots. Under 3-2-1, the teams that finish ninth or tenth get more chances in the drawing than those that finish seventh or eighth. A team in that range therefore improves its chances by finishing lower [7]. The new lottery also sets its odds by the current season alone, so a team that has been weak for many years gets no extra help.

The impossibility result depends on measuring weakness by a single season's standings. We introduce Carry-Over Lottery Allocation (COLA), which measures weakness by playoff results over many seasons. Each team has a draft priority, a score that carries over from one season to the next and gives the method its name. A team's priority grows each season it goes without playoff success. It falls when the team wins a playoff series, which is one round of the playoffs between two teams, or when it receives a top pick. COLA works like a waiting list. One more loss this season leaves a team's place on the list unchanged, so a draft that follows the list need not reward losing. Help still reaches weak teams, because the teams that stay weak are the ones that wait longest.

COLA is a family of rules for setting the draft order, and we call each such rule a mechanism. Full-lottery COLA holds a drawing for every pick among teams that miss the playoffs, and a team's chances are proportional to its priority. Simple COLA has no drawing. The team that has waited longest picks first. The NBA's 3-2-1 can be read as an extreme member of the family. This season's standings alone set its priority, and it carries only one thing from year to year: a limit on how often the same team can win the top picks. The same priority could also be used in an auction, a different kind of draft in which teams bid for players or for draft positions. It would decide how much each team has to bid.

We focus on Simple COLA. For each team, it counts the seasons since the team last won a playoff series or received one of the first three draft picks. The team with the highest count picks first. If two teams have the same count, the one with more wins this season picks ahead of the other. When no team can improve its pick by losing on purpose, a mechanism is called incentive compatible. We prove that Simple COLA is incentive compatible. The proof rests on a few assumptions about how teams behave, and all but one are standard. The exception, which Section 4 defends, is that a team would rather win a game than lose it to change which other teams it competes with for the top picks.

We test nine mechanisms, including the NBA's old lottery and 3-2-1, in ZenGM Basketball-GM. It is an open-source simulator, meaning its code is public, and it simulates every game and every change to each team's roster in realistic fictional leagues. Each mechanism runs on the same 30 leagues for 25 seasons. We compare the mechanisms on three measures. The first is how much they help the teams that have waited the longest. The second is how evenly playoff success spreads across teams over many seasons, which we call long-term parity. The third is the reward for losing, meaning how much a team's pick improves when it loses more games. Because every mechanism plays the same 30 leagues, we can compare any two of them league by league. For each comparison, we report a 95% confidence interval, the range that very likely contains the true difference. The leagues are fictional, so we also check whether the reward for losing and the help for long-waiting teams look the same on real NBA standings, one season at a time.

Of the nine mechanisms, Simple COLA does the most for the teams that have waited longest. Under 3-2-1, the longest any team goes without a playoff series win or one of the first three picks averages 16.3 seasons. Under Simple COLA, it averages 5.5. No team that misses the playoffs ever gains by losing more games. The average wait for a team's next playoff series win is within about a tenth of a season of the wait under 3-2-1. Simple COLA has two costs, which Sections 5 and 7 measure. First, teams wait about 0.2 seasons longer to return to the playoffs. Second, a team that has waited a long time can gain by losing a first-round playoff series. A series win would send it to the back of the list. Every number in this paper can be reproduced with the code in our public repository. The 3-2-1 lottery expires after the 2029 draft unless the league votes to keep it [7], so the league will soon choose a draft mechanism again.

## 2. Background

Teams have tanked more in the years when the draft paid more for losing. Until 1984, the last-place teams in both conferences flipped a coin for the first pick, and the other teams picked in reverse order of record. For the next five drafts, a lottery gave every team that missed the playoffs the same chance at the first pick, a format we call equal odds. In 1990, the NBA weighted the lottery so a worse record brought a better chance [6]. Taylor and Trogdon [5] compared three seasons, one played under each format. They looked at teams that could no longer reach the playoffs, which we call eliminated teams. They found tanking by eliminated teams under reverse order and under the weighted lottery, and none under equal odds. Price et al. [8] extended the test to every game from 1977 to 2007 and found more late-season losing in the seasons when finishing last paid the most. Bookmakers set their odds as if eliminated teams tank, although the same study finds mixed evidence of tanking in the actual scores [9]. Fornwagner [10] finds that National Hockey League teams also lose more after elimination.

Banchio and Munro's impossibility result explains why equal odds were the exception, and its proof needs only two teams [6]. Take two eliminated teams with the same record that meet in the last game of the season. If finishing lower gives a better chance at the first pick, the losing team comes out ahead, so neither team wants to win. A lottery based on the final standings can favor the worst teams only by giving a lower finish a better chance, so it always creates this problem. Equal odds avoids the problem by favoring no team. The NBA abandoned equal odds after five drafts because owners felt it did not help the worst teams [9]. The 3-2-1 lottery, which the league's owners approved 29 to 1 on May 28, 2026, starts with the 2027 draft and moves back toward equal odds [11]. It expands the lottery from fourteen teams to sixteen, with each team getting one, two, or three chances in the drawing. The drawing decides all sixteen picks, whereas the old lottery decided only the first four [7].

Most other proposals still judge teams on a single season and change which of its games count. Banchio and Munro would update the lottery odds after every game, then stop changing them when a team could first gain by losing [6]. We call that point the freeze. Applied to the 1988-89 season, their rule freezes the odds after the 332nd of 1,025 games and gives last-place Miami a 45.5% chance at the first pick. The odds then reflect how teams played before the freeze, regardless of what happened later. Charlotte finished with the second-worst record, yet it would have had a 7.5% chance. Indiana won eight more games than Charlotte and would have had an 18.5% chance, because more of Charlotte's losses came after the freeze. Kazachkov and Vardi [12] would rank the teams that miss the playoffs by where they stood at a point in the season fixed in advance. Lenten et al. [13] would give the first pick to the team eliminated earliest, and Gold [14] would give it to the team that wins the most games after it is eliminated. Banchio and Munro note that neither rule is fully incentive compatible [6].

A last group drops the lottery. The wheel, designed by Mike Zarren of the Boston Celtics, fixes each team's draft position years in advance. Over thirty years, each team picks first once, second once, and so on down to thirtieth [15], [16]. Losing buys nothing, and a weak team gets no extra help. Two other plans use an auction. Nate Silver [17] would replace the draft with player bidding. Each team would get between 25 and 100 units to bid with each year, and the worst records would get the most. A Boston Celtics proposal reported by Quinn [18] would have teams bid credits for draft positions. Unspent credits carry over to the next year, and a team loses 20 credits for each playoff series it wins.

None of the draft proposals above look beyond the current season to decide which teams need help. The Celtics' credits do, because they carry over and shrink when a team wins in the playoffs, but they work through an auction. COLA uses the same longer view to set the draft order directly, without an auction.

# References


[1] ESPN, "Utah Jazz at Orlando Magic, February 7, 2026: Box score," 2026. [Online]. Available: https://www.espn.com/nba/boxscore/_/gameId/401810605

[2] NBA, "NBA fines Jazz and Pacers," NBA Communications, Feb. 12, 2026. [Online]. Available: https://pr.nba.com/nba-fines-jazz-and-pacers/

[3] The Ringer Staff, "The NBA's Tank Off Is Here: Who Is the Most Shameless Loser?" The Ringer, Feb. 10, 2026. [Online]. Available: https://www.theringer.com/2026/02/10/nba/nba-teams-tanking-2026-draft-lottery-odds

[4] NBA, "NBA Board of Governors approves changes to Draft Lottery system," NBA Communications, Sep. 28, 2017. [Online]. Available: https://pr.nba.com/nba-draft-lottery-reform-player-rest-rules

[5] B. A. Taylor and J. G. Trogdon, "Losing to win: Tournament incentives in the National Basketball Association," *Journal of Labor Economics*, vol. 20, no. 1, pp. 23–41, 2002.

[6] M. Banchio and E. Munro, "A no-tanking draft allocation policy," in *MIT Sloan Sports Analytics Conference*, 2020. [Online]. Available: https://www.sloansportsconference.com/research-papers/a-no-tanking-draft-allocation-policy

[7] K. O'Connor, "The NBA's 3-2-1 lottery system is here. How will it impact future drafts?" Yahoo Sports, May 29, 2026. [Online]. Available: https://sports.yahoo.com/nba/article/the-nbas-3-2-1-lottery-system-is-here-how-will-it-impact-future-drafts-175414282.html

[8] J. Price, B. P. Soebbing, D. Berri, and B. R. Humphreys, "Tournament incentives, league policy, and NBA team performance revisited," *Journal of Sports Economics*, vol. 11, no. 2, pp. 117–135, 2010.

[9] B. P. Soebbing and B. R. Humphreys, "Do gamblers think that teams tank? Evidence from the NBA," *Contemporary Economic Policy*, vol. 31, no. 2, pp. 301–313, 2013.

[10] H. Fornwagner, "Incentives to lose revisited: The NHL and its tournament incentives," *Journal of Economic Psychology*, vol. 75, p. 102088, 2019.

[11] T. Bontemps, "NBA's BOG votes to expand lottery, addresses tanking problem," ESPN, May 28, 2026. [Online]. Available: https://www.espn.com/nba/story/_/id/48903380/sources-nba-bog-votes-expand-lottery-disincentivizes-tanking

[12] A. M. Kazachkov and S. Vardi, "On tanking and competitive balance: Reconciling conflicting incentives," Working paper, Mar. 2020. [Online]. Available: https://akazachk.github.io/pubs/tanking.pdf

[13] L. J. A. Lenten, A. C. T. Smith, and N. Boys, "Evaluating an alternative draft pick allocation policy to reduce 'tanking' in the Australian Football League," *European Journal of Operational Research*, vol. 267, no. 1, pp. 315–320, 2018.

[14] A. M. Gold, "NHL draft order based on mathematical elimination," *Journal of Quantitative Analysis in Sports*, vol. 6, no. 1, 2010.

[15] Z. Lowe, "The NBA's Possible Solution for Tanking: Good-bye to the Lottery, Hello to the Wheel," Grantland, Dec. 23, 2013. [Online]. Available: https://grantland.com/the-triangle/the-nbas-possible-solution-for-tanking-good-bye-to-the-lottery-hello-to-the-wheel/

[16] ——, “Wheel of Misfortune,” Grantland, Mar. 18, 2014. [Online]. Available: https://grantland.com/features/wheel-of-misfortune/

[17] N. Silver, “Our radical plan to replace the NBA draft,” Silver Bulletin, Apr. 1, 2026. [Online]. Available: https://www.natesilver.net/p/radical-plan-to-replace-the-nba-draft-lottery-arc-auction

[18] S. Quinn, “Goodbye NBA Draft Lottery? Explaining ‘draft credit’ system that the league could start using as soon as 2030,” CBS Sports, May 29, 2026. [Online]. Available: https://www.cbssports.com/nba/news/goodbye-nba-draft-lottery-explaining-draft-credit-system/